

SIMATS ENGINEERING

CO- 2: AT – 2: Simulation Task - Medium

COURSE NAME: Advance Wireless Communication Systems /
5G / 6G Communication Systems
COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 2 - Analyse LTE and 5G wireless network architectures, including carrier aggregation, MIMO techniques, beamforming strategies, and service-based core network functional entities. (BL4)

Assessment Tool Weightage: 35%

Each student should simulate one question and should upload the results in Github account along with the necessary outputs and codes. This assessment tool is evaluated for 50 marks.

QUESTIONS:	Total Marks: 50	Duration: 90 Mins
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1. Simulation of interference management under user density, reuse factor, SNR, SINR, and throughput (BL4)
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Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding Model Design	20%	Demonstrates strong understanding of underlying concepts in the simulation 5	Shows good understanding with minor gaps 4	Partial understanding; some misconceptions 3	Lacks understanding of key concepts 1-2
Tool Usage	20%	Uses simulation tools effectively with correct setup and execution 5	Uses tools with minor errors or guidance 4	Limited ability to use tools properly 3	Unable to use simulation tools 1-2
Analysis of Results	20%	Provides clear, logical analysis and meaningful interpretation of results 5	Analysis is reasonable with minor gaps 4	Superficial analysis; limited interpretation 3	Unable to analyze or interpret results 1-2
Documentation	20%	Presents results clearly with proper documentation 5	Documentation is adequate with minor issues 4	Poorly organized or incomplete documentation 3	No proper documentation or presentation 1-2

Criteria	Question Number :			
	Level 4	Level 3	Level 2	Level 1
Conceptual Understanding Model Design				
Tool Usage				
Analysis of Results				
Documentation				

Code:

```
clc;
clear;
close all;

%% Parameters
user_density = 10:10:200;    % Number of users
reuse_factor = [1 3 7];     % Frequency reuse factors
SNR_dB = 25;                 % Fixed Signal-to-Noise Ratio

figure;

for k = 1:length(reuse_factor)

    R = reuse_factor(k);

    % Simple Interference Model
    interference = (user_density/20) ./ R;

    % SINR Calculation
    SINR_dB = SNR_dB - 10*log10(1 + interference);

    % Throughput using Shannon Capacity
    SINR_linear = 10.^(SINR_dB/10);
    Bandwidth = 10;          % MHz

    Throughput = Bandwidth .* log2(1 + SINR_linear);

    %% Graph 1
    subplot(2,2,1)
    hold on
    plot(user_density,SINR_dB,'LineWidth',2)
    xlabel('User Density')
    ylabel('SINR (dB)')
    title('User Density vs SINR')
    grid on

    %% Graph 2
    subplot(2,2,2)
    hold on
    plot(user_density,Throughput,'LineWidth',2)
    xlabel('User Density')
    ylabel('Throughput (Mbps)')
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        title('User Density vs Throughput')
        grid on

    end

    legend('Reuse = 1','Reuse = 3','Reuse = 7')

%% Graph 3
    subplot(2,2,3)

    SNR = 0:2:40;

    interference = 2;

    SINR = SNR - 10*log10(1+interference);

    plot(SNR,SINR,'r','LineWidth',2)

    xlabel('SNR (dB)')
    ylabel('SINR (dB)')
    title('SNR vs SINR')
    grid on

%% Graph 4

    subplot(2,2,4)

    reuse = 1:7;

    Throughput2 = zeros(size(reuse));

    for i=1:length(reuse)

        inter = 5/reuse(i);

        sinr = SNR_dB - 10*log10(1+inter);

        Throughput2(i)=Bandwidth*log2(1+10^(sinr/10));

    end

    plot(reuse,Throughput2,'ko-','LineWidth',2)

    xlabel('Reuse Factor')

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ylabel('Throughput (Mbps)')
title('Reuse Factor vs Throughput')
grid on

```

Output:

